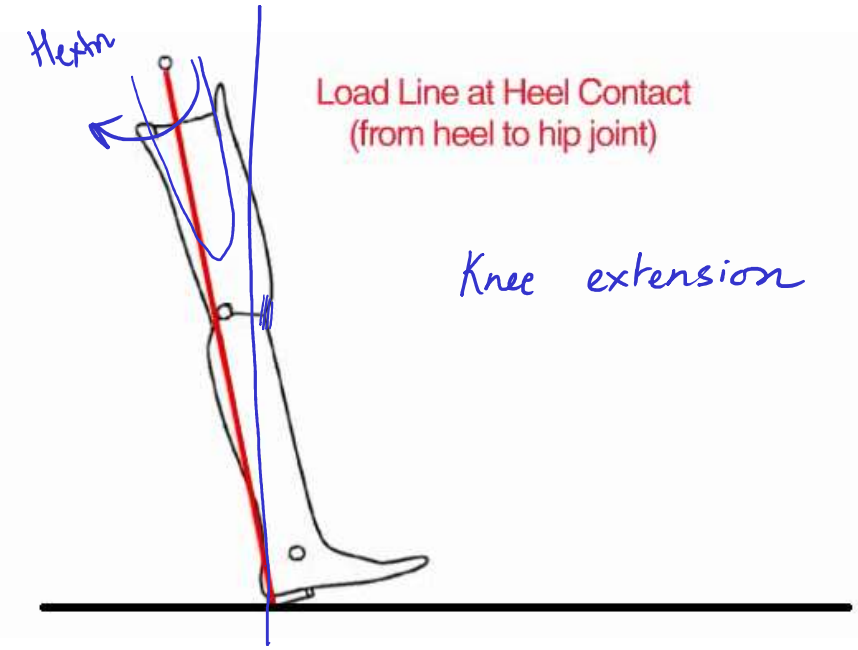
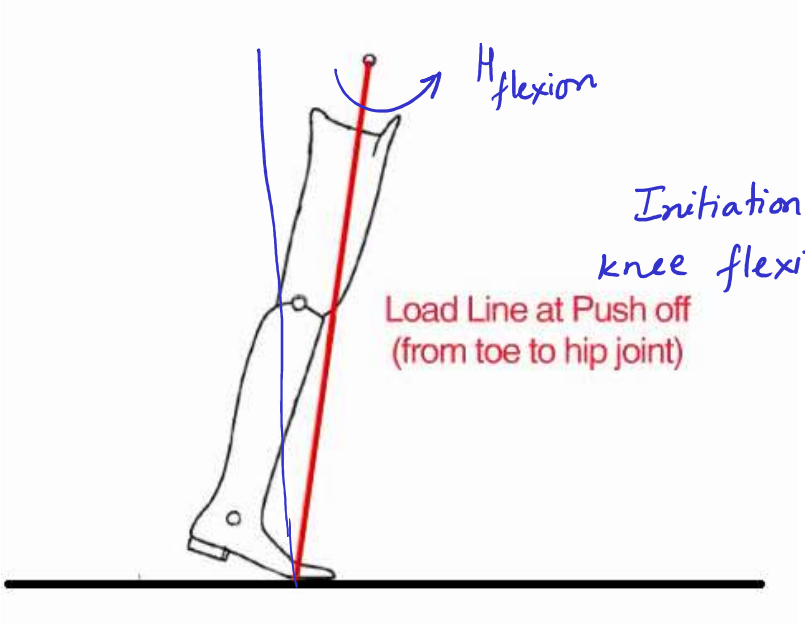
Walking with a single-axis knee



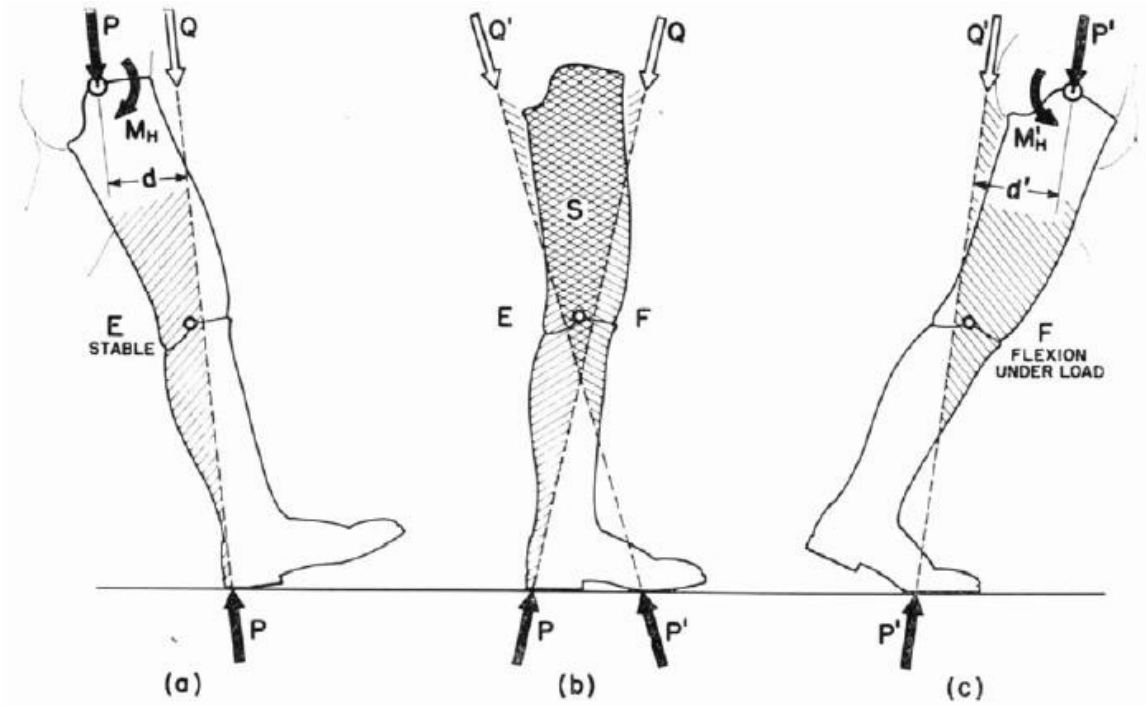
Stability at heel strike



Initiation of swing

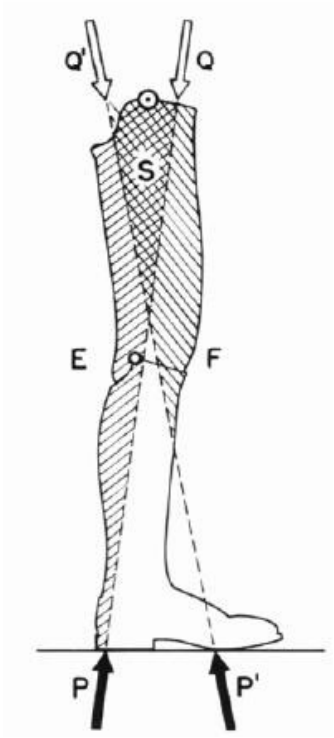
Animation courtesy TS Anand, Alumnus R2D2, IITM

Knee stability



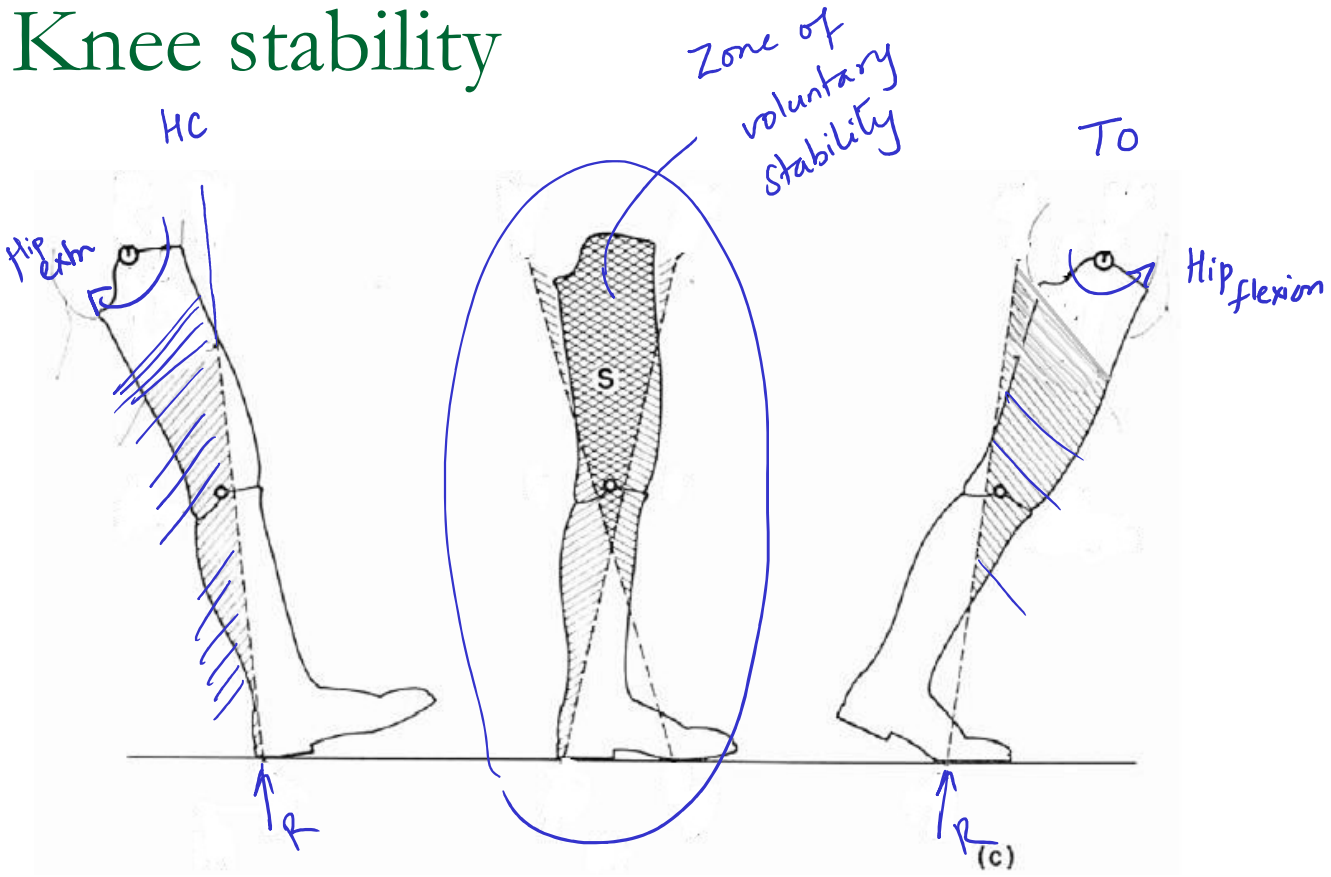
Zone of voluntary control of knee stability

Figure courtesy Radcliffe, 1977 – Above Knee Prosthetics, THE KNUD JANSEN LECTURE



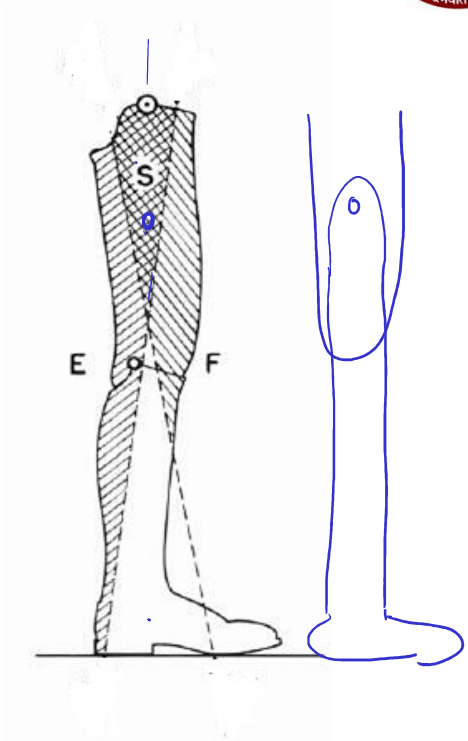
Zone for person with impaired hip musculature due to atrophy/amputation

Knee stability



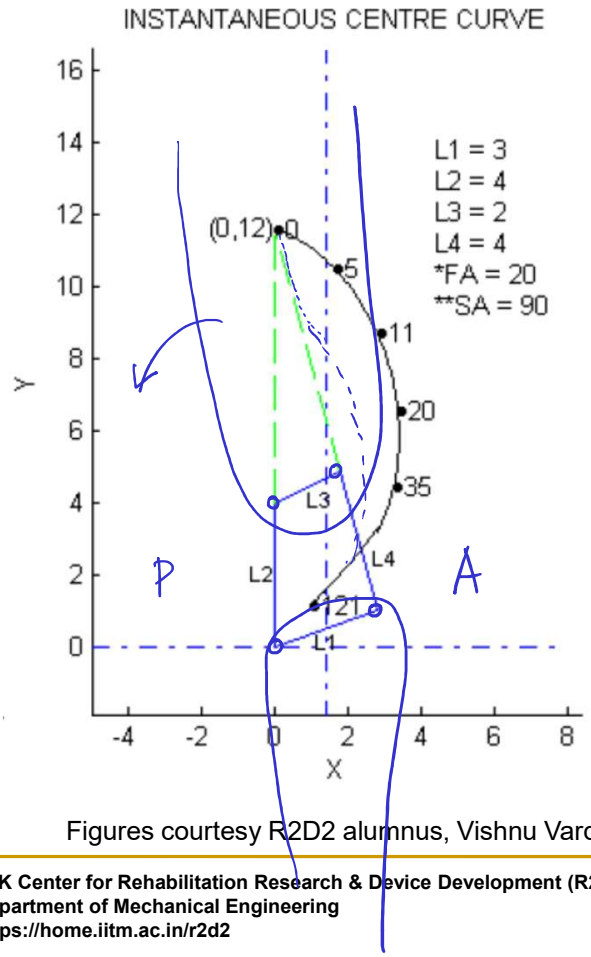
Zone of voluntary control of knee stability

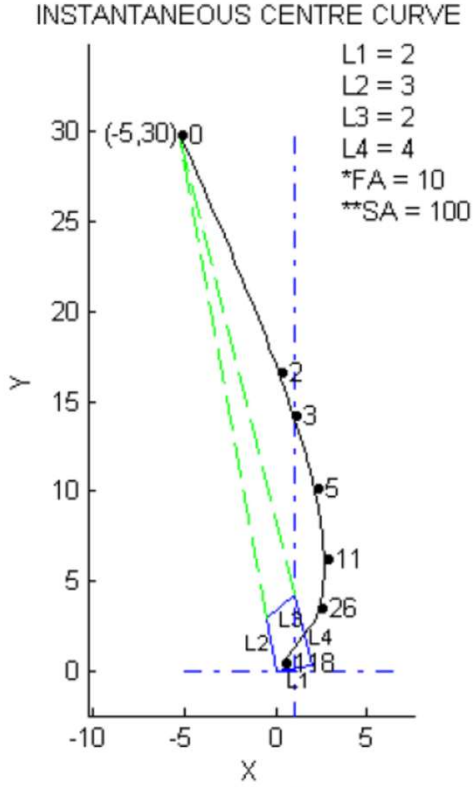
Figure adapted from Radcliffe, 1977 – Above Knee Prosthetics, THE KNUD JANSEN LECTURE

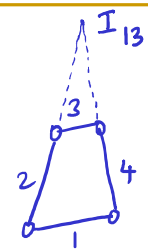


Zone for person with impaired hip musculature due to atrophy/amputation

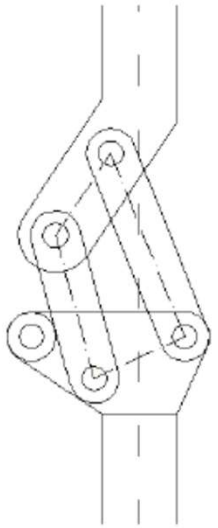
Polycentric knees

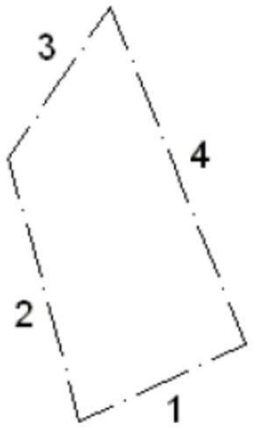






(instant centre of rotn of link 3 w.r.t link 1





FA refers to angle of fixed link with horizontal

SA refers to the starting angle of link 2 with horizontal in CCW direction

Figures courtesy R2D2 alumnus, Vishnu Vardhan Reddy (M.Tech, 2009)

Polycentric knees

Inverted slider-crank



Ossur Polycentric Knee



OB 3R78



Geoflex® Knee

Polycentric knees

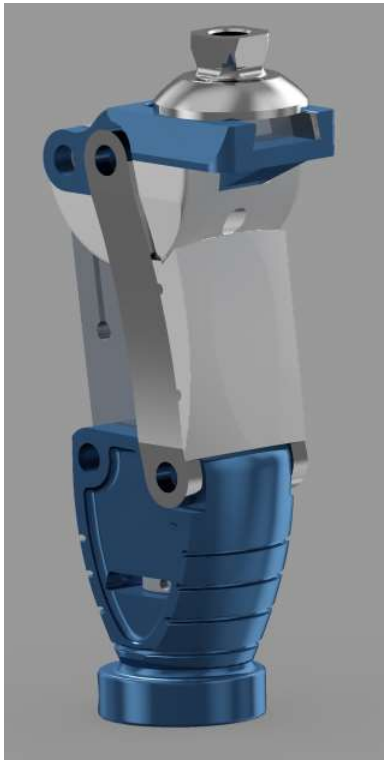


JaipurKnee



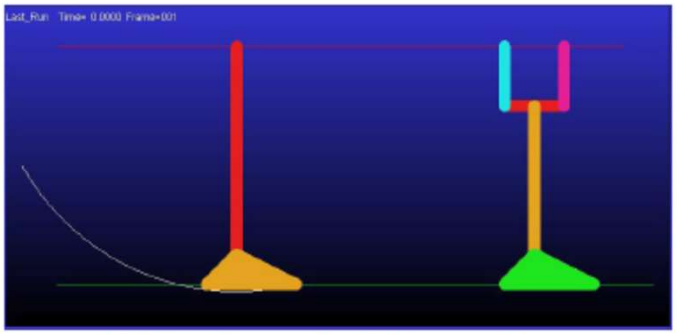
ReMotion Knee



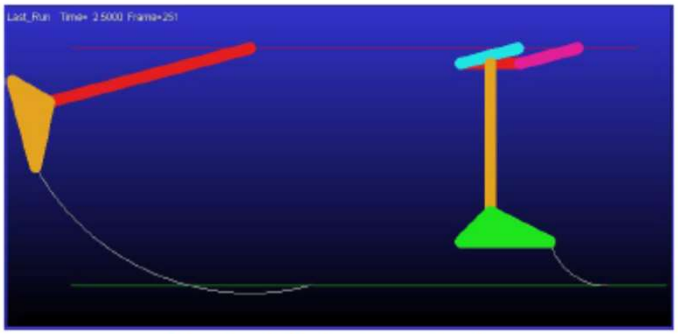


IITM R2D2 Knee

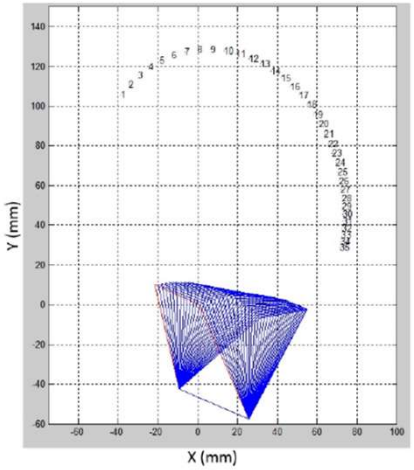
Polycentric knee



A Hinge knee and a Polycentric knee



Toe loci during knee flexion





IITM R2D2 Knee v1

Courtesy TS Anand, Alumnus R2D2, IITM

Polycentric knee – toe clearance

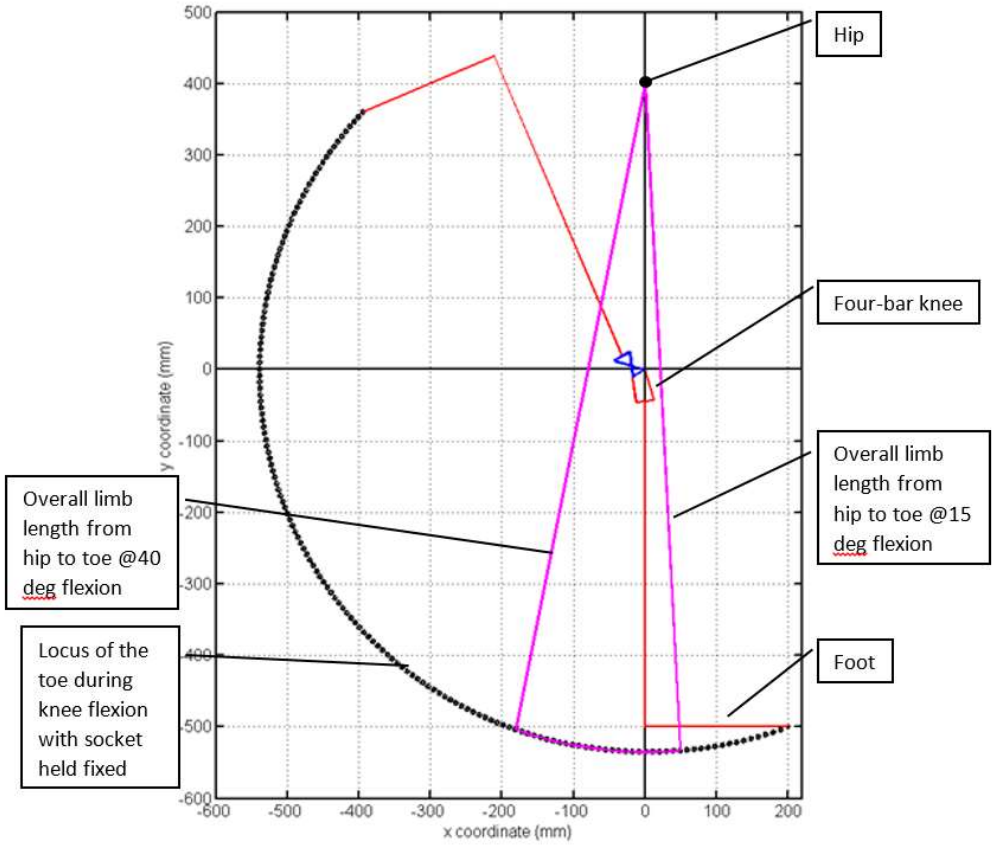


Figure 4. Illustration of toe clearance calculation of a typical four-bar knee

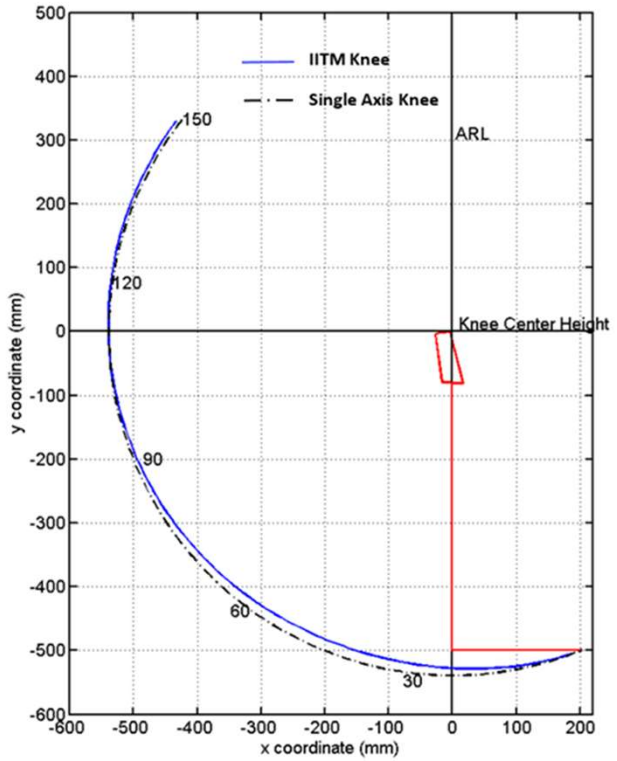
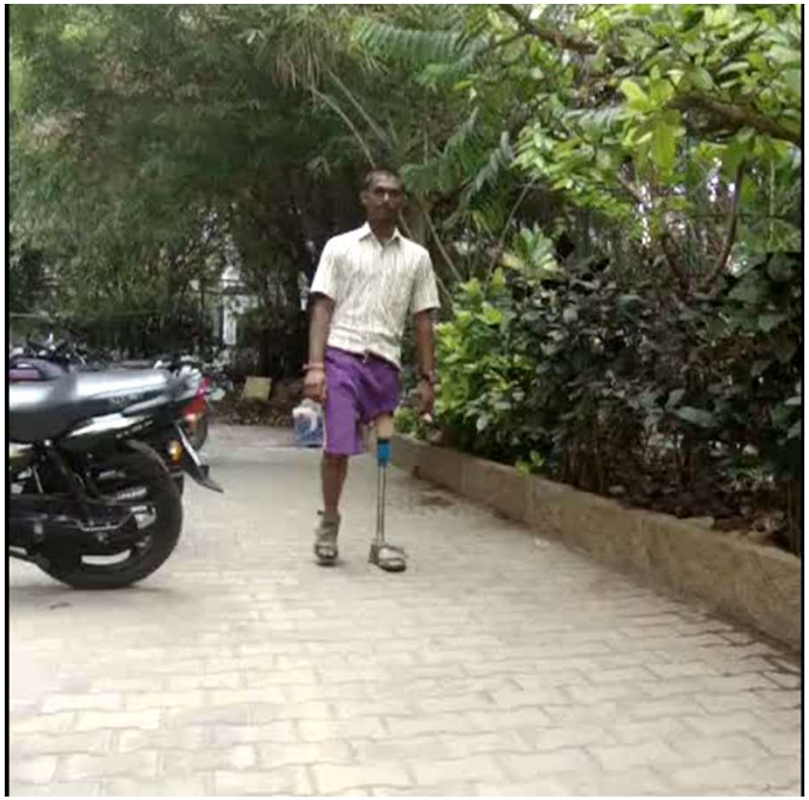


Figure 10: Loci of the toe with knee flexion. The numbering on the curves specify the knee flexion angle. (All dimensions in mm)

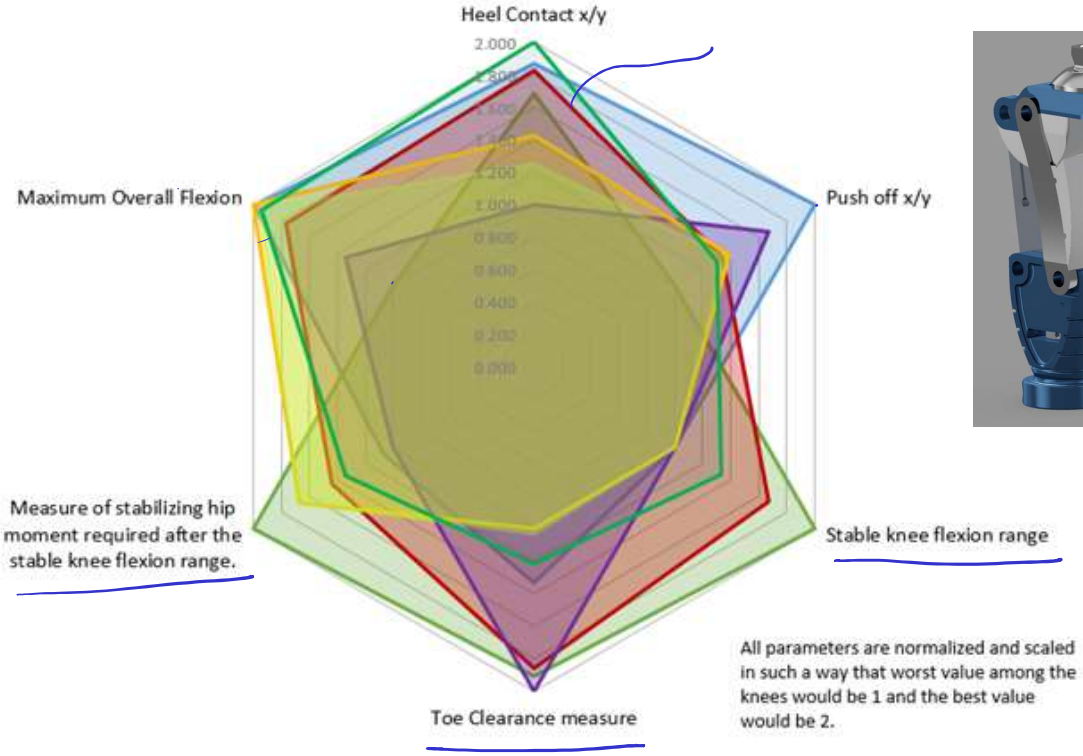
Courtesy TS Anand, Alumnus R2D2, IITM (M.Tech 2014)

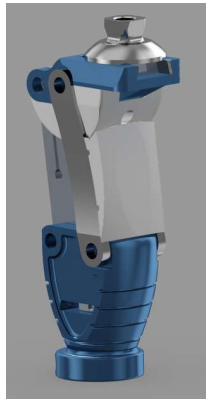
IITM Knee



Knee Trial at Mobility India







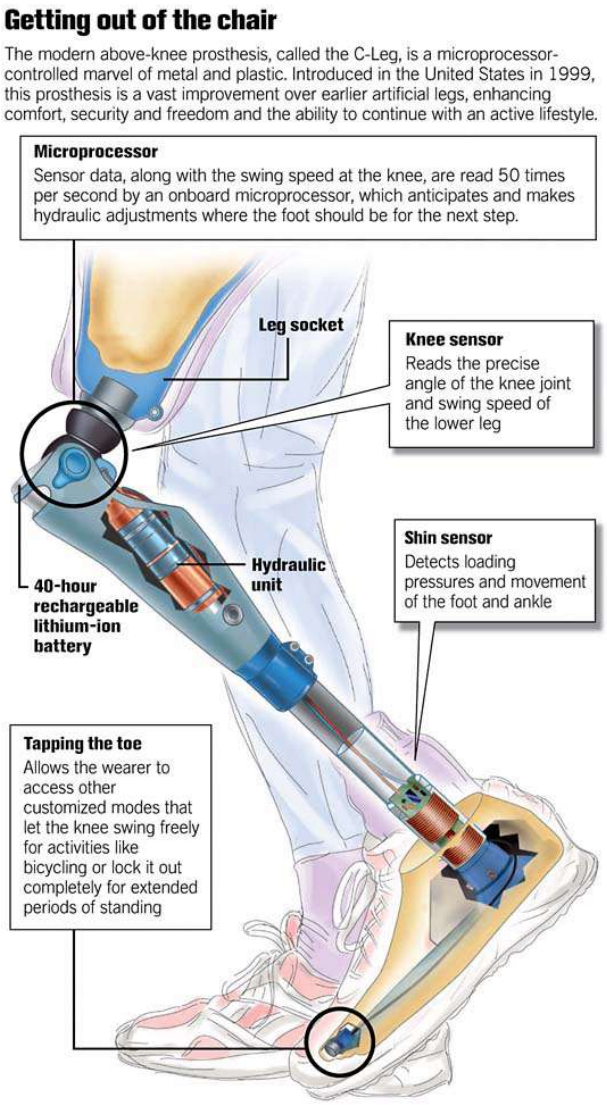
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Department of Mechanical Engineering
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T. S. Anand and S. Sujatha. A Method for Performance Comparison of Polycentric Knees and its Application to the Design of a Knee for Developing Countries. Prosthetics and Orthotics International, 41(4), 402--411, 2017.

Prosthetic knees



Otto Bock C-Leg



http://www.sissonmobility.com/c-leg/